

SIMATS ENGINEERING

CSA1511 – CLOUD COMPUTING AND BIG DATA ANALYTICS

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Cloud Computing Application Based Questions Report

Application Based Question 1

Question:

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

Aim: To explain and analyze the given cloud computing application scenario.

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Key Points

- Scalability - Important aspect discussed for this application.
- High Availability - Important aspect discussed for this application.
- Security - Important aspect discussed for this application.
- Cost Optimization - Important aspect discussed for this application.
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- Reliability - Important aspect discussed for this application.

Real-World Example Example: Large cloud platforms such as Amazon, Netflix, banking systems and global mobile applications use these cloud technologies to provide uninterrupted services to millions of users. Example: Large cloud platforms such as Amazon, Netflix, banking systems and global mobile applications use these cloud technologies to provide uninterrupted services to millions of users.

Advantages

- Improves performance
- Reduces downtime
- Enhances security
- Supports global users
- Reduces operational cost

Conclusion The discussed cloud solution effectively addresses the business requirement by providing scalability, reliability, security and better performance. The discussed cloud solution effectively addresses the business requirement by providing scalability, reliability, security and better performance.

References AWS Documentation, Microsoft Azure Documentation, Google Cloud Documentation.

Application Based Question 2

Question:

2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy.

How can encryption and access control protect sensitive backup data?

Explain the cost advantages compared to traditional storage systems.

Provide an example of a disaster recovery use case.

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Explanation Cloud services improve scalability, availability, security, cost efficiency and performance. Relevant technologies such as auto-scaling, load balancing, cloud storage, CDNs, compliance mechanisms, containers and Kubernetes are applied depending on the scenario. These services ensure efficient resource utilization, business continuity, fault tolerance and better user experience. Cloud services improve scalability, availability, security, cost efficiency and performance. Relevant technologies such as auto-scaling, load balancing, cloud storage, CDNs, compliance mechanisms, containers and Kubernetes are applied depending on the scenario. These services ensure efficient resource utilization, business continuity, fault tolerance and better user experience. Cloud services improve scalability, availability, security, cost efficiency and performance. Relevant technologies such as auto-scaling, load balancing, cloud storage, CDNs, compliance mechanisms, containers and Kubernetes are applied depending on the

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Application Based Question 3

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3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

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Application Based Question 4

Question:

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.

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Application Based Question 5

Question:

5. A development team wants to adopt containerization for application deployment.

Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments.

Provide an example of a microservices-based application.

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